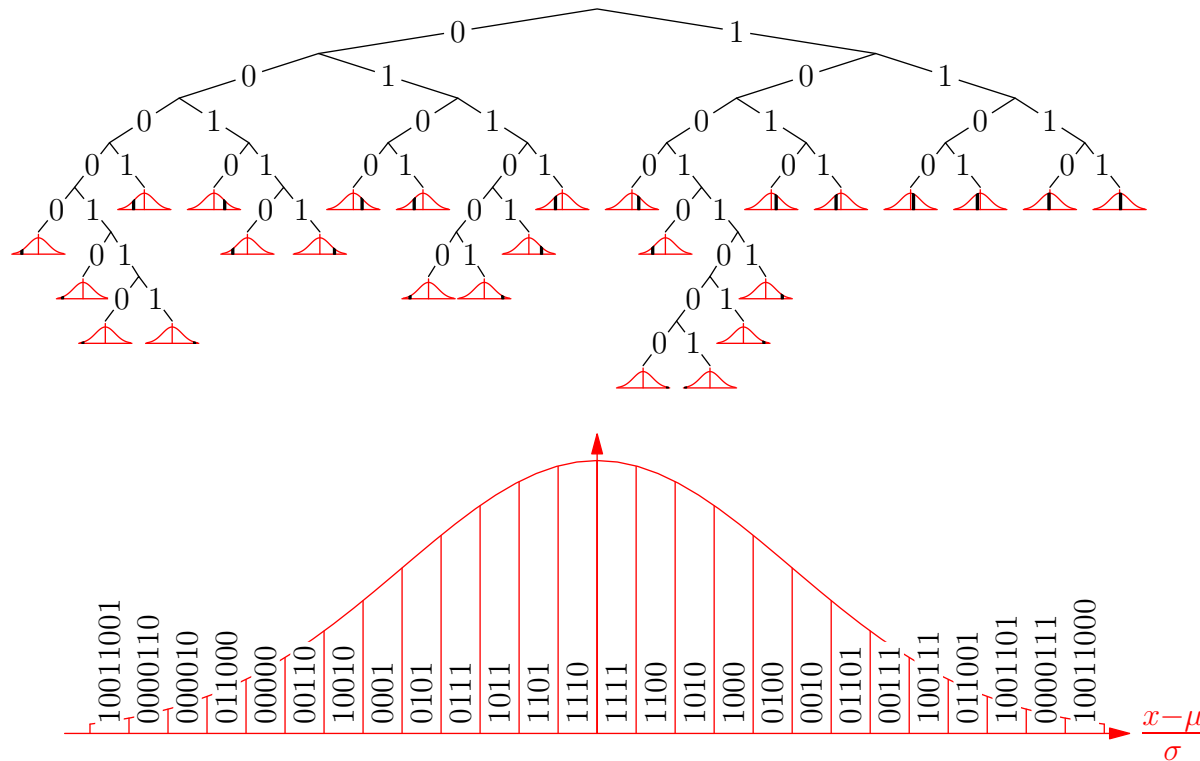


Mathematics of Machine Learning

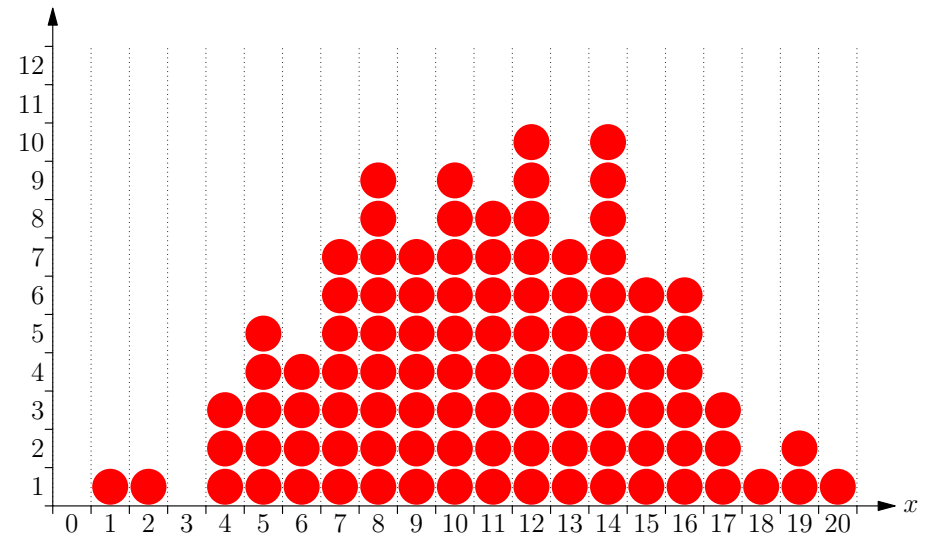
Entropy



Entropy, Coding, Maximum Entropy

Outline

1. **Measuring Uncertainty**
2. Code Length
3. Maximum Entropy



Measuring Uncertainty

- What is more uncertain tossing a coin three times or throwing a dice
- The answer depends on whether you care about the order of the coin tosses
- But, how do we answer such a question?
- Let X be a random variable denoting the possible outcomes
- Interestingly, Shannon entropy give a precise answer

$$H_X = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x))$$

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Let's Calculate

- For an honest dice $D \in \{1,2,3,4,5,6\}$ and $\mathbb{P}(D = i) = 1/6$ so

$$H_D = -\sum_{i=1}^6 \frac{1}{6} \log_2 \left(\frac{1}{6} \right) = -\log_2 \left(\frac{1}{6} \right) = \log_2(6) \approx 2.584 \text{ bits}$$

- For an honest coin where we care about the order so $C \in \{000,001,\dots,111\}$ the $\mathbb{P}(C = i) = \frac{1}{8}$ and

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Unordered Coin Toss

- What if we don't care about the order of the out-come then $\mathbb{P}(HHH) = \mathbb{P}(TTT) = 1/8$, $\mathbb{P}(HHT) = \mathbb{P}(HTT) = 3/8$ so

$$H_U = -\frac{1}{4}\log_2\left(\frac{1}{8}\right) - \frac{3}{4}\log_2\left(\frac{3}{8}\right) \approx 1.811 \text{ bits}$$

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- But why Shannon entropy?

Additive Entropy

- If H_X and H_Y is the uncertainty of two independent random variable X and Y , what is the uncertainty of the combined event (X,Y) ?

$$\begin{aligned} H_{(X,Y)} &= - \sum_{X,Y} \mathbb{P}(X,Y) \log_2(\mathbb{P}(X,Y)) \\ &= - \sum_{X,Y} \mathbb{P}(X)\mathbb{P}(Y) \log_2(\mathbb{P}(X)\mathbb{P}(Y)) \\ &= - \sum_{X,Y} \mathbb{P}(X)\mathbb{P}(Y) (\log_2(\mathbb{P}(X)) + \log_2(\mathbb{P}(Y))) \\ &= - \sum_X \mathbb{P}(X) \log_2(\mathbb{P}(X)) - \sum_Y \mathbb{P}(Y) \log_2(Y) = H_X + H_Y \end{aligned}$$

- Shannon's entropy is one of the few functions that satisfy this condition

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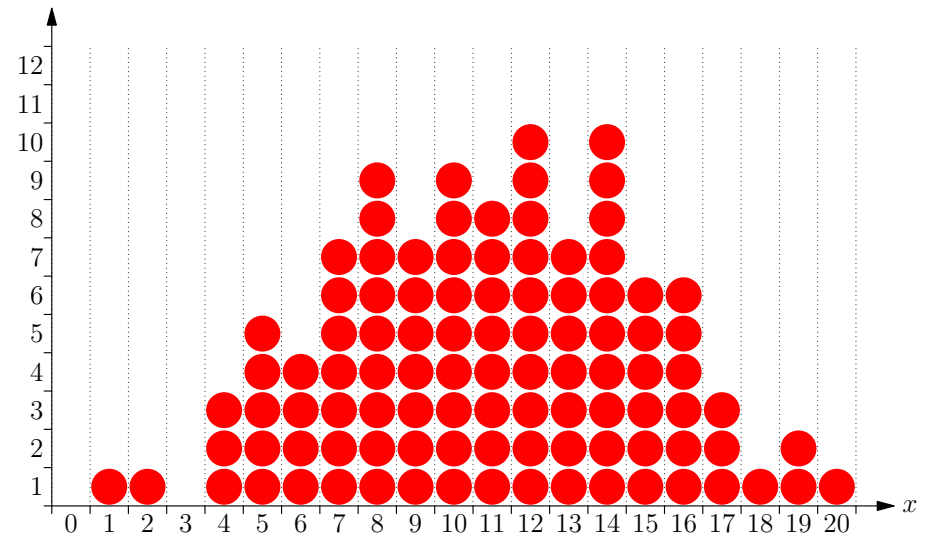
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Why Measure Entropy in Bits

- Suppose we had to communicate a message with 2^n equally likely outputs (e.g. the result of n -coin tosses)
- We can do this with a binary string with n bits (011..0)
- If there were 5 possible outcomes I could do this with 3 bits, but waste $3/8$ of the message
- However if we have a batch of 3 independent messages each with 5 outcomes then there are 125 possible outcomes
- By batching together enough messages with N outcomes then we asymptotically need just $\log_2(N)$ bits

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Different Probabilities

- We “showed” that if we had N events, X_i , each with probability, $\mathbb{P}(X_i) = 1/N$, we can code the outcomes with a message of length $-\log_2(\mathbb{P}(X_i))$
- With a shorter message we would not be able to distinguish all possible outcomes from the message
- What happens if some of outcomes occur with a different probability

X_i :	1	2	3	4	5	6
$p(X_i)$:	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$
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- If the probabilities are not equal to 2^{-n} we can still find a code with a length very close to $-\log_2(\mathbb{P}(X))$ per message by transmitting a large number of messages
- The length of the message measures the amount of **surprise** on receiving the message
- Shannon's entropy is the expected length of the message to communicate a random variable X

$$H_X = \mathbb{E}_X[-\log_2(\mathbb{P}(X))] = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x))$$

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- The expected length is a measure of the uncertainty (how much information on average we need to convey the outcome)

Real Codes

- Those of you with some computer science background will realise that we can't actually use different length strings in a code without paying some price
- We won't know where a code word ends so we can't decode the message
- An optimal solution is to use Huffman encoding where we associate the leaf of a tree with each code word
- Using the tree we can decode any message constructed using the tree
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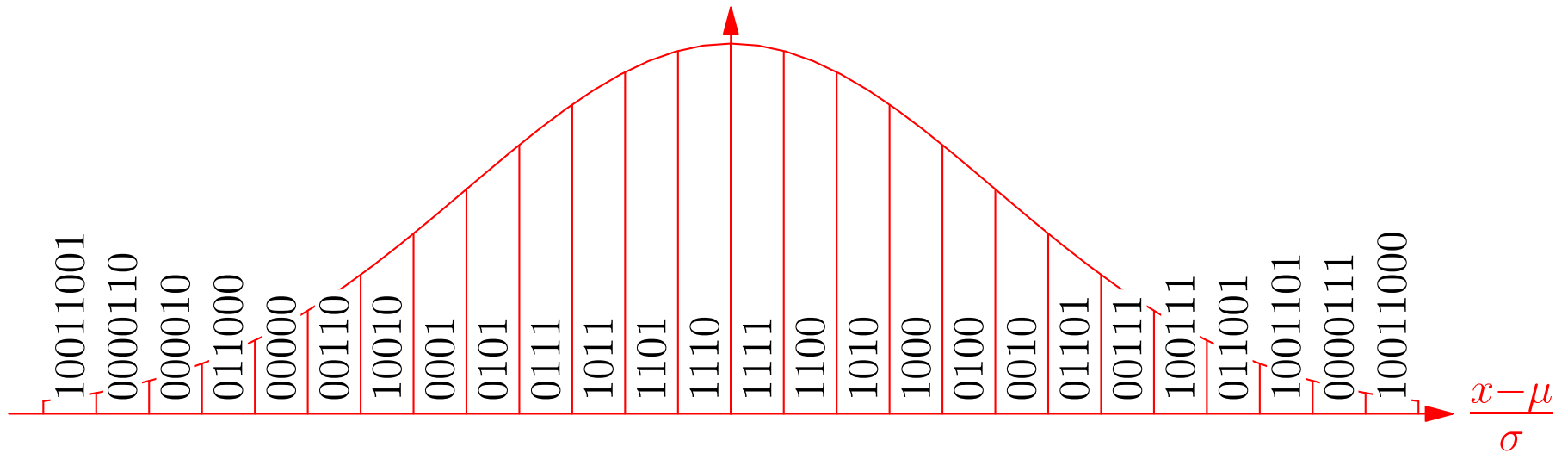
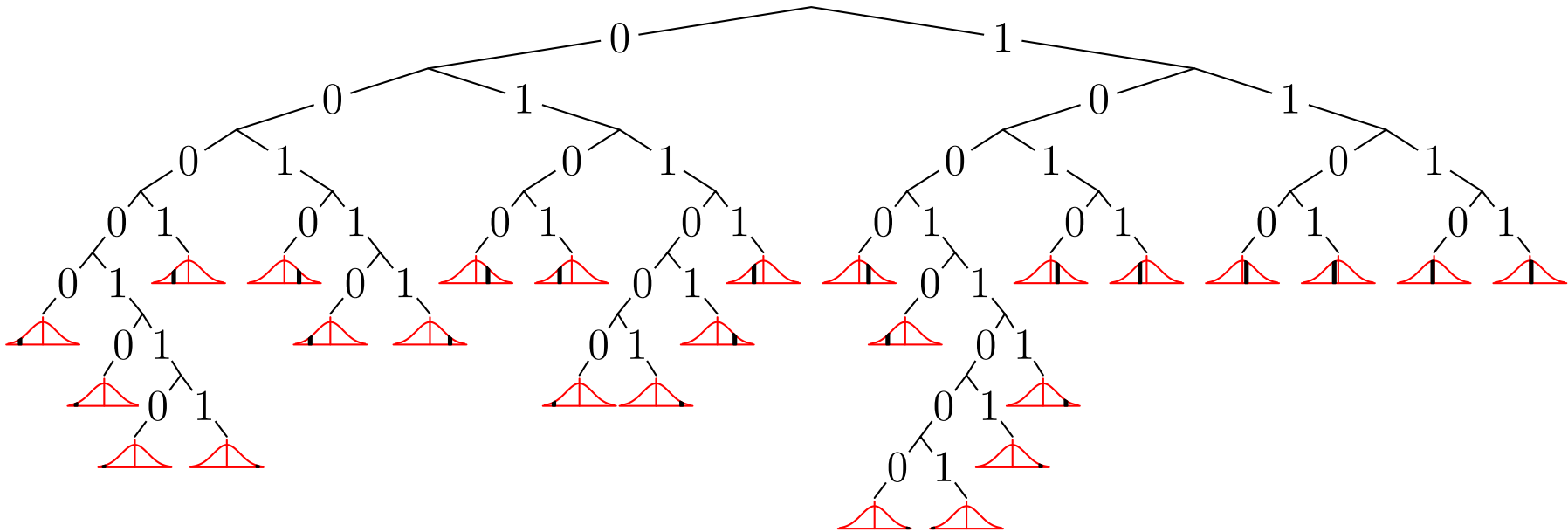
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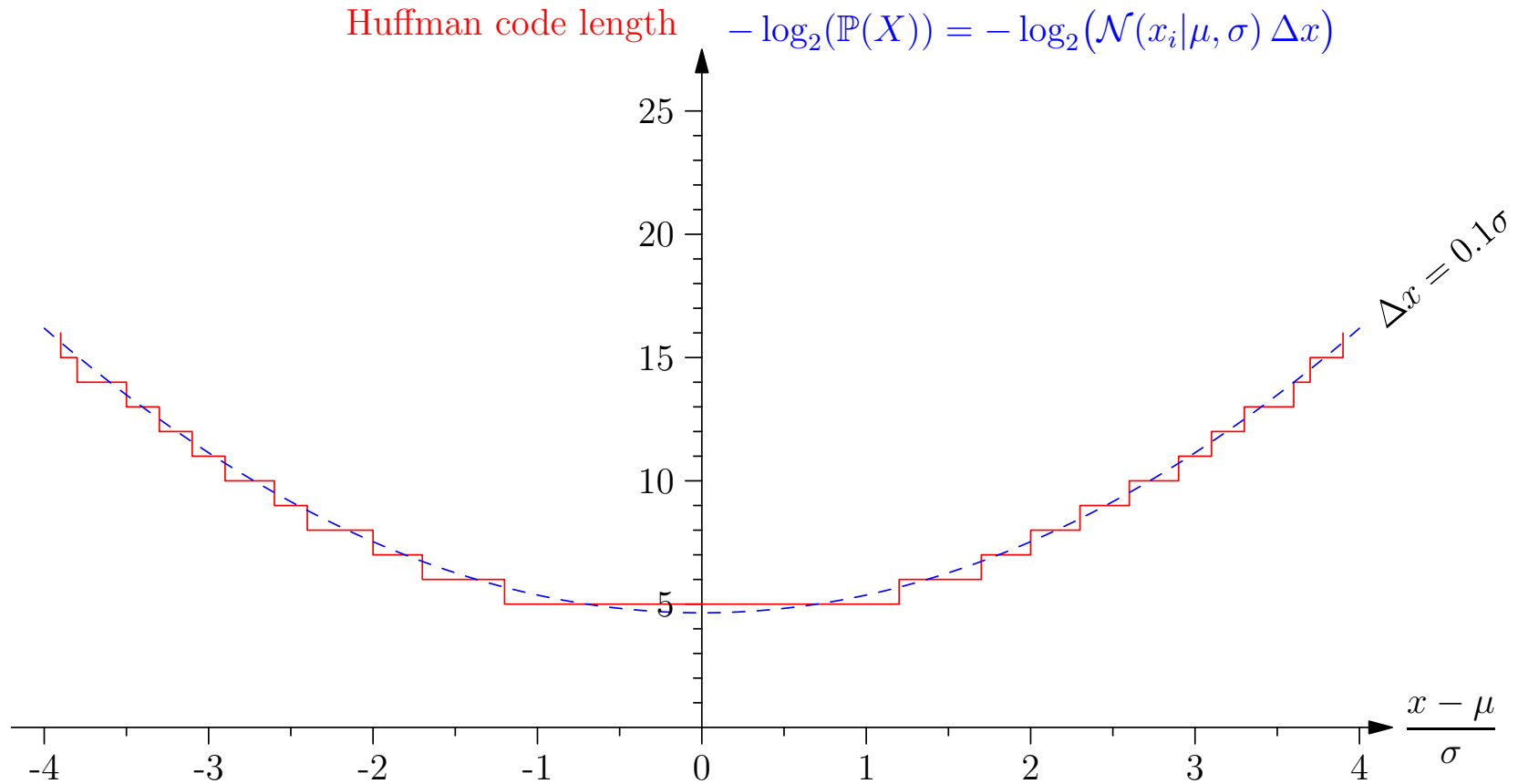
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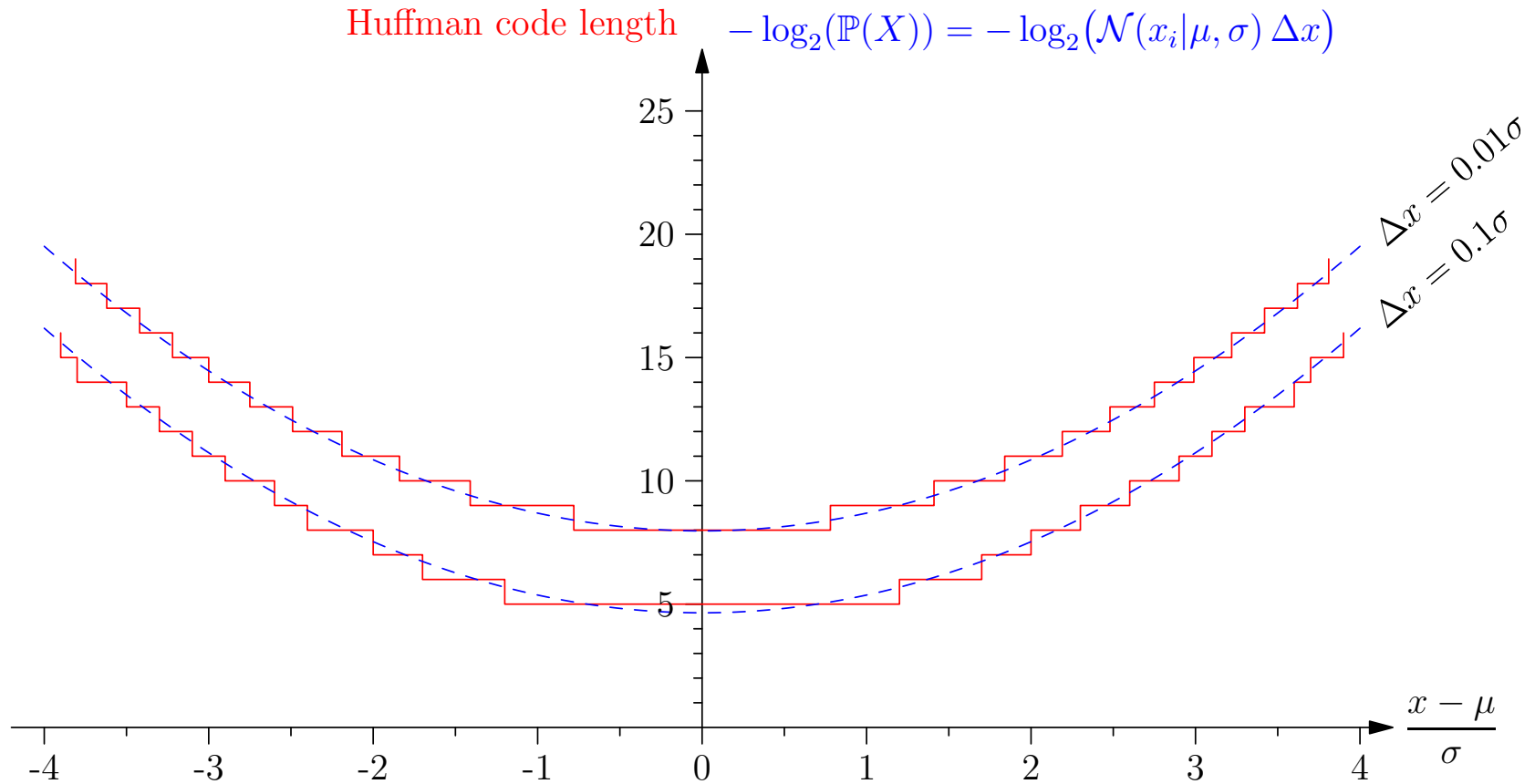
Coding Normals



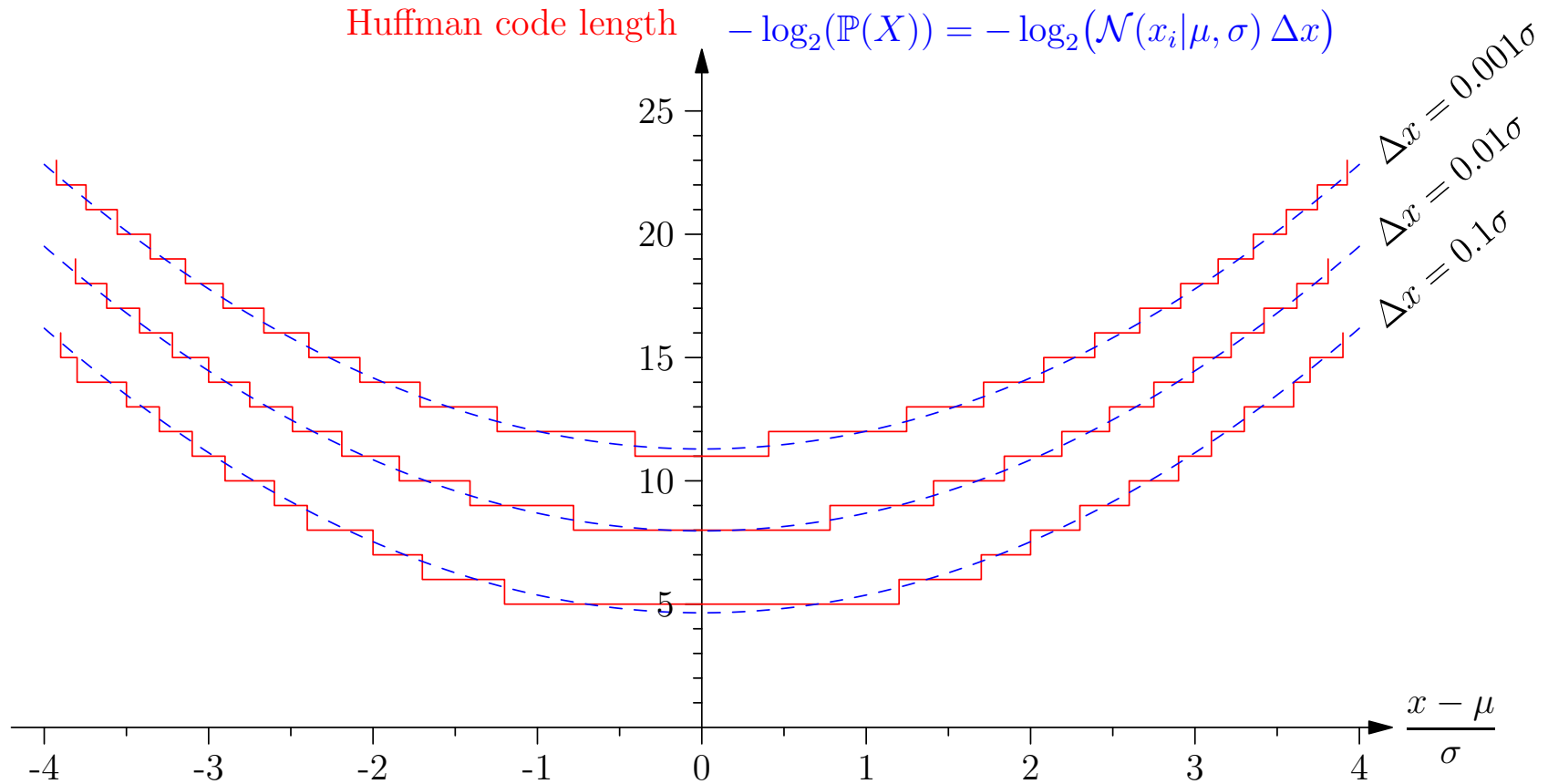
Coding Normals to Accuracy Δx



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bits and nats

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$$H_X = - \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) \log_2(\mathbb{P}(X = x))$$

- Sometimes it is easier to use natural logarithms

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- In this case the entropy is measured in **nats** with 1 nat equal to $\log_2(e)$ bits
- This is often easier when we want to do calculus on entropy

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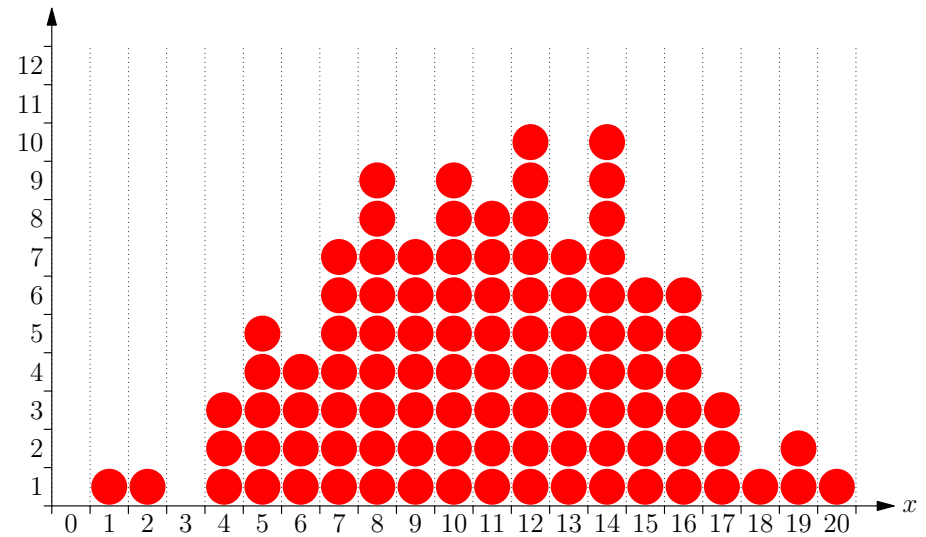
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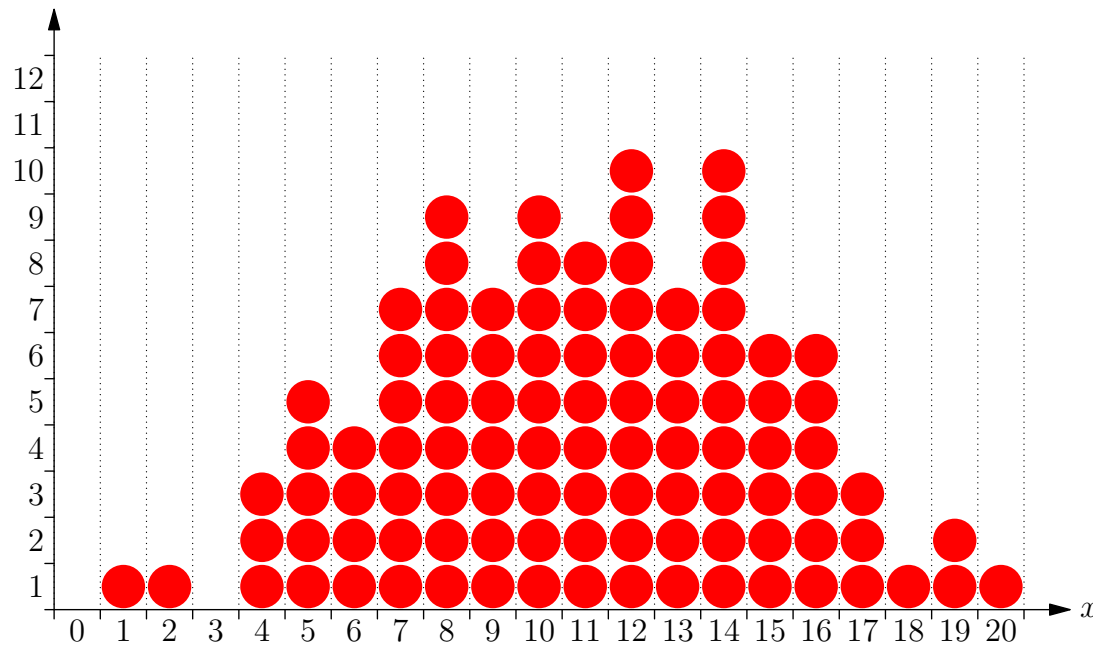
Outline

1. Measuring Uncertainty
2. Code Length
3. **Maximum Entropy**



Number of States

- Suppose I have N balls I then put in K boxes so that n_i balls finish up in box i



- Counting the number of ways to do this

$$\mathbb{P}(\mathbf{n}) = \frac{N!}{n_1!n_2!\cdots n_K!}$$

Stirling's Approximation

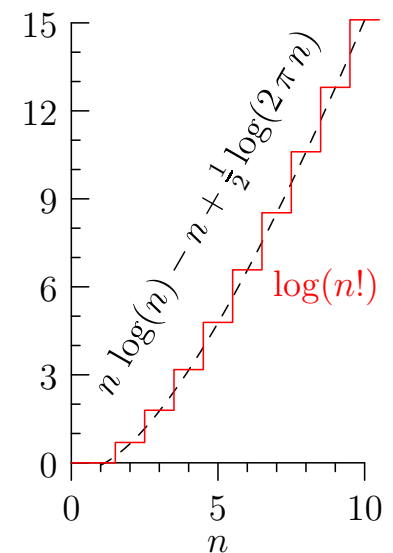
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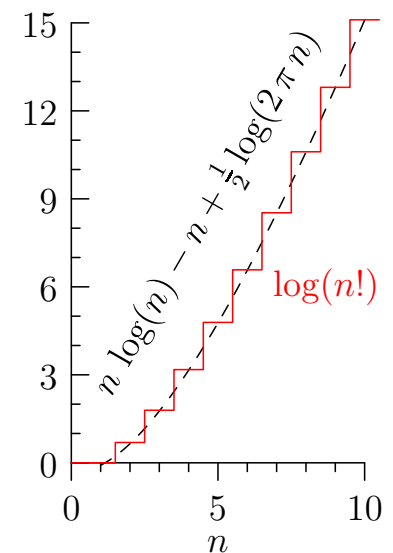
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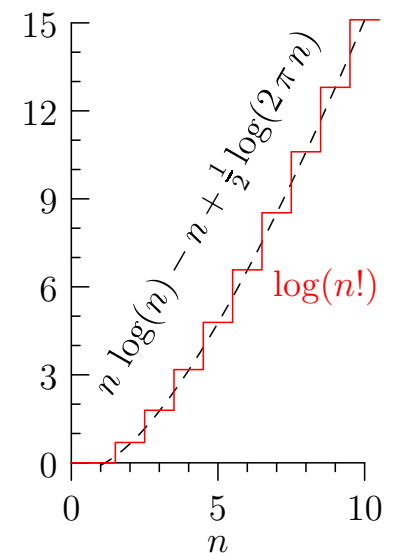
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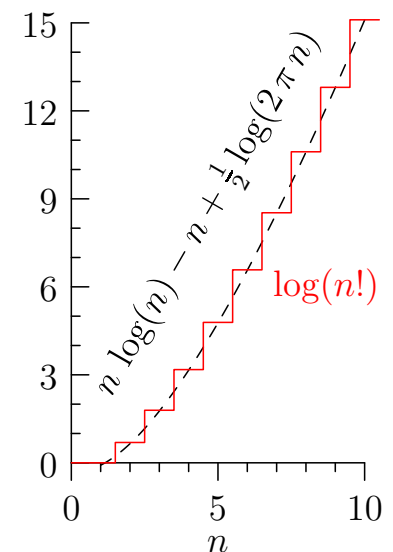
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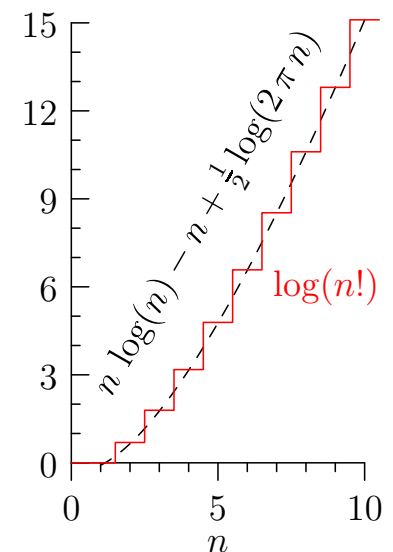
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Maximum Entropy Method

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- This is known as the **maximum entropy method**
- We can rationalise this as this is by far the most likely set of configurations consistent with the observations
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- Consider a continuous random variable, X , with a known mean and second moment

$$\mathbb{E}[X] = \mu, \quad \mathbb{E}[X^2] = \mu_2 = \mu^2 + \sigma^2$$

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- The place that they work superbly well is in statistical physics
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